

Rahul Kumar

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EDUCATION

Indian Institute of Information Technology, Design and Manufacturing

Madhya Pradesh, India

Bachelor of Technology – Computer Science and Engineering

Aug 2024 – Aug 2028

- Courses: Operating Systems, Data Structures, Analysis Of Algorithms, Artificial Intelligence, Machine Learning, Networking, Databases

SKILLS SUMMARY

Languages: Python, PHP, C++, C, JavaScript, SQL, JAVA, HTML, CSS, TypeScript

Frameworks: React, Redux, Tailwind CSS, Scikit, TensorFlow, Django, Node.js, Express.js, Next.js

Tools: Pandas, NumPy, Matplotlib, GIT, PostgreSQL, MySQL, Jupyter

Databases: MongoDB, MySQL

Platforms: Web, Windows, AWS

Soft Skills: Leadership, Event Management, Writing, Public Speaking, Time Management

PROJECTS

AI Notes Generator | *Next.js, BullMQ, Redis, MongoDB, Gemini AI* | [Live Link](#)

2026

- Developed an enterprise-grade AI study platform that automatically generates structured notes, interactive MCQs, and Mermaid flowcharts from user prompts and PDF documents.
- Architected a scalable background processing system using **BullMQ** and **Upstash Redis** to decouple heavy AI workloads, completely bypassing serverless timeouts.
- Engineered a robust multi-cloud deployment across **Vercel** (Frontend) and **Render** (Background Worker) to handle concurrent user requests with zero downtime.
- Implemented rate-limiting, **Clerk** authentication, and Vector-based **RAG** (Retrieval-Augmented Generation) to allow users to interactively chat with their generated notes.
- Designed a premium glassmorphism UI with **Tailwind CSS**, featuring real-time job progress tracking, study analytics, and 1-click WhatsApp public sharing.

Unified CP Tracker | *PHP, MySQL*

2025

- Orchestrated the integration of APIs from Codeforces, CodeChef, and LeetCode into a singular dashboard; enhanced the user experience, leading to a 25 percent increase in user engagement on the platform.
- Designed contest insights, performance visualization, and secure authentication for better learning outcomes.

ACHIEVEMENTS

Competitive Programming Profiles

- Codeforces – 70+ Problems Solved, Max Rating: 1133 (Newbie)
- CodeChef – Max Rating: 1453 (2 Star)
 - Best Global Rank: 509 in CodeChef Starters 223
- LeetCode – 300+ Problems Solved
 - Max Day Streak: 50+

HACKBYTE 3.0 SPONSOR TEAM

Hackbyte 3.0 Sponsor Team

2024

IIT Jabalpur

- Wrote several proposals and emails, securing sponsors, which led to the event being 3x bigger than the previous year.